CNN-based Region-of-Interest Image Reconstruction from Truncated Data in Cone-Beam CT

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ABSTRACT

We developed a region-of-interest (ROI) image reconstruction method that effectively reduces truncation artifacts in CBCT. By using U-Net-based deep learning (DL) methods, we devised a method to reduce truncation artifacts for ROI imaging. A total of 16294 image slices from 49 patient cases were used to generate projection data. The center of the projected image was cropped to a width of 150 mm. Then, the outer part of the truncation image was filled with each outermost pixel value for the initial correction. After the filtering process, the truncation area was cut off and used as input data in the DL model. Finally, inference images were reconstructed by use of the FDK algorithm. SSIM values for the test set of 14 patients were calculated as 0.541, 0.709 and 0.979 for FBP, Extension and the proposed ROI method, respectively. We have achieved promising results and believe that the proposed ROI image reconstruction method can help reduce radiation dose while preserving image quality.

Keywords: X-ray, Convolutional neural network, Truncation artifact, ROI image, Low-dose, CBCT

1. INTRODUCTION

Diagnostic computed tomography (CT) is a technique that uses radiation to express anatomical information in the human body as a three-dimensional image, and is contributing greatly to improving the diagnosis and treatment effect of diseases in clinical practice. Cone-beam CT (CBCT) was widely used in angiography, radiotherapy, and various surgeries for determining the current position of the target and the surrounding normal tissues and confirmation of the results of surgery.

Despite such usefulness and utility, the frequency of use is limited due to the side effects of radiation on the human body. Therefore, a reduction of radiation dose is the most important factor in CBCT techniques or applications. Accordingly, various studies are being conducted to reduce the amount of radiation exposed to the patient while maintaining the quality of the image[1].

Recent technologies related to CBCT image reconstruction and restoration to reduce radiation exposure include iterative reconstruction techniques, machine learning and deep learning methodologies, and region of interest (ROI) CT imaging techniques. Iterative reconstruction (IR) methods formulate the mathematical inverse problem in CT imaging regularization techniques to reconstruct data that has been collected with sparse projection data or low exposures. If the image is reconstructed using the conventional filtered back projection (FBP), the image quality is noisy or streak artifacts occur. IR studies have contributed greatly to dose reduction in this respect. However, there is a burden of imaging time due to iterative calculations, and in order to secure diagnostic performance, it is also necessary to tune according to the data. On the other hand, machine and deep learning (DL) methods have shown higher accuracy and speed improvement in image reconstruction. It is being used for high-quality, low-dose CT imaging using various methods such as supervised learning, unsupervised learning, and generative adversarial network method.

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Interior computed tomography collects ROI data by using only a limited field of view (FOV) from sinogram or projection data, so the tissues and organs surrounding the ROI are not exposed to radiation as much as the collimated region[2]. This technique is called truncated projection, interior tomography, or ROI imaging (Figure 1). This sampling strategy is useful for imaging protocols where data from surrounding structures are not essential for diagnostic tasks.

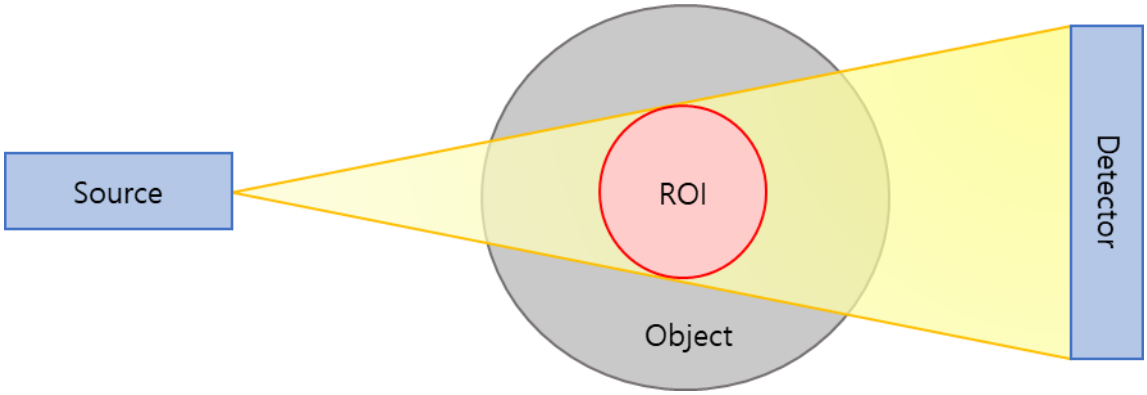


Figure 1. Demonstrates the concept of ROI imaging technology.

However, when the truncated projection data is reconstructed using the FBP, bright-rim artifacts (truncation artifacts) occur in the outer portion of the FOV. To solve this problem, a technique for optimizing filters, iterative reconstruction methods such as a total variation (TV), and a deep learning method for reducing artifacts based on reconstructed images have been proposed[2-5].

In this study, we developed a ROI image reconstruction method that effectively reduces truncation artifacts in CBCT and reduces radiation exposure to the patient. By using U-Net-based DL methods, we devised a method to reduce truncation artifact in filtrated projection domain for ROI imaging.

2. METHOD

The dataset consisted of clinical CT images from a chest CT dataset (AAPM grand challenge)[6]. A total of 16294 image slices from 49 patient cases were used to generate projection data.

The CT data set showed a slightly different body size for each patient case due to the difference in pixel pitch. Therefore, as shown in Figure 2, we tried to reduce the bias in performance by classifying the CT data set so that the patient's body size is similar in the training set and the test set.

Figure 2 shows the process of preparing the ground truth (GT) data and learning data set of this study. To acquire the projection images, a standard ray-driven numerical forward projection procedure was performed in full scan mode implemented internally. To generate truncated data to use for training, first the center of the projected image was cropped to a width of 150 mm. Then, the initial correction was performed by applying a simple method of mirroring the outermost value of the existing truncation data. That is, the outer part of the truncation image was filled with each outermost pixel value for the initial correction of the truncation artifact. After the filtering process, the truncation artifact was created and the truncation area was cut off. These cut-off data used as input data in the DL model (Figure 3). We used the widely known U-Net as the backbone network to improve the quality of images with truncation artifact. The network training was performed on a Nvidia RTX A6000 GPU. The network was trained using Adam optimizer with a L1 loss function. No additional regularizer was added to the loss function. The dataset for training and test consisted of 10894 and 5400 images, respectively. A total of 250 epochs training were performed. Finally, inference images were reconstructed by use of the Feldkamp–Davis–Kress (FDK) algorithm.

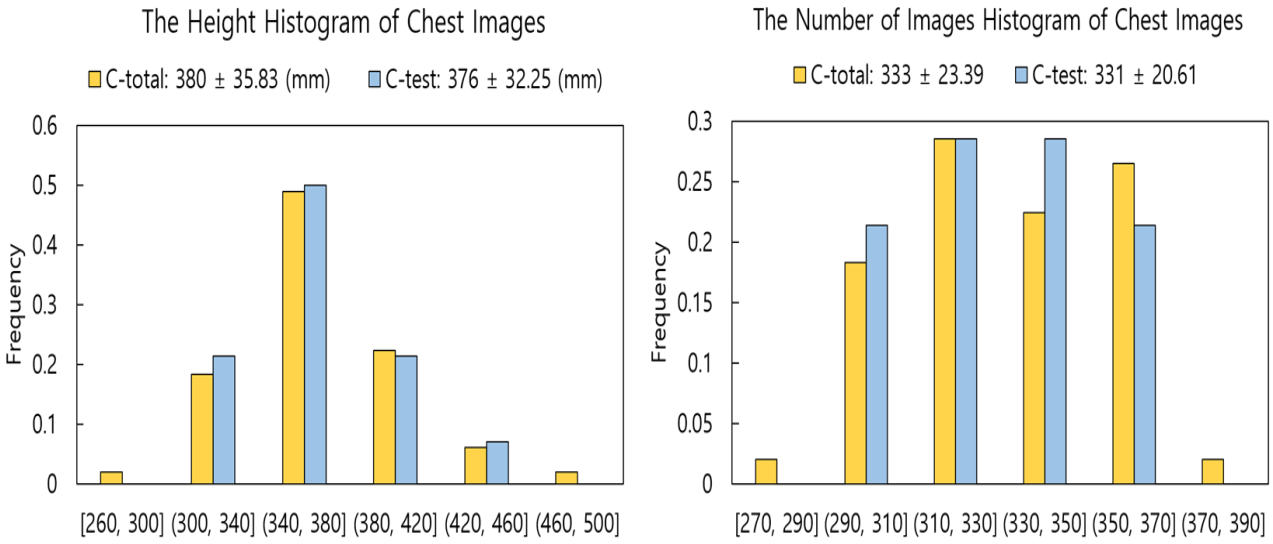


Figure 2. This figure shows the distribution according to the size of the data set used in this study.

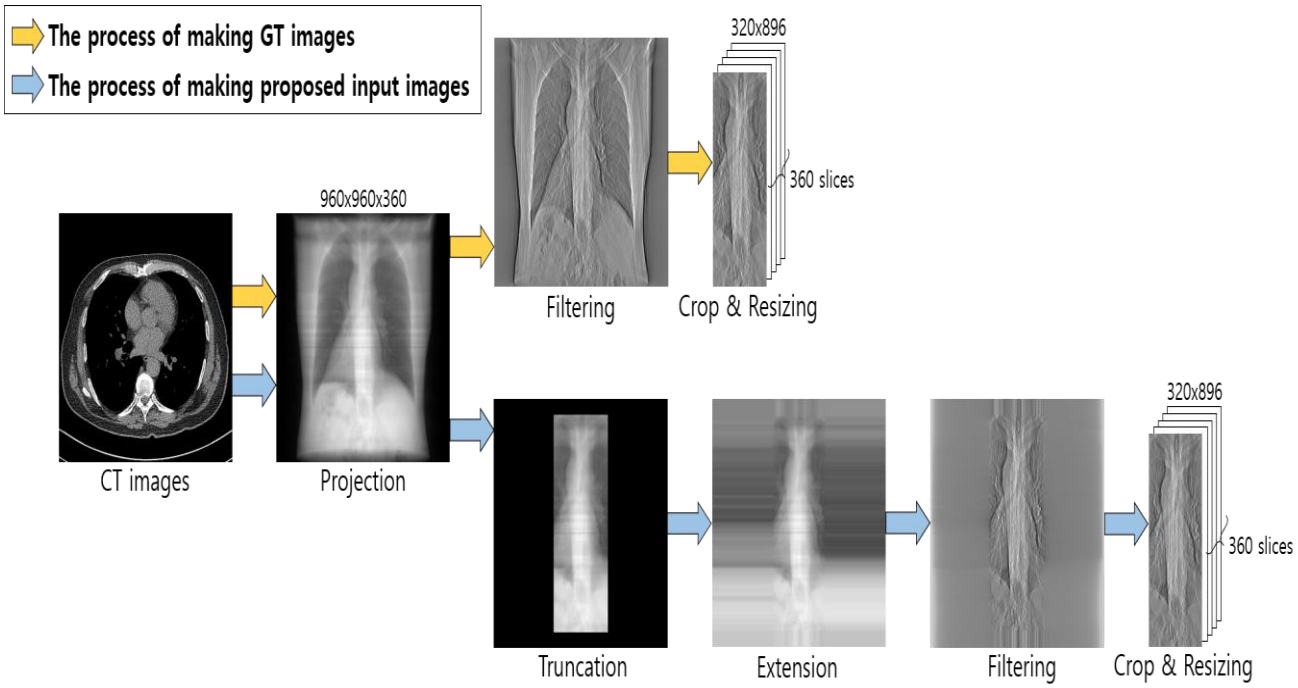


Figure 3. The procedure of data preparation. The yellow arrow represents the process of making Ground Truth (GT) images, and the blue arrow represents the process of making input images for DL input.

3. RESULT

Figure 4 shows projection images corrected by GT, filtered projection, extension projection, and the proposed method, respectively. It can be seen that the truncation artifact is greatly generated in the filtered projection, and it can be seen that the truncation artifact is relatively decreased in the extension projection image. However, it is shown that there are fine streak artifacts in the overall vertical direction and the noise of the image increases.

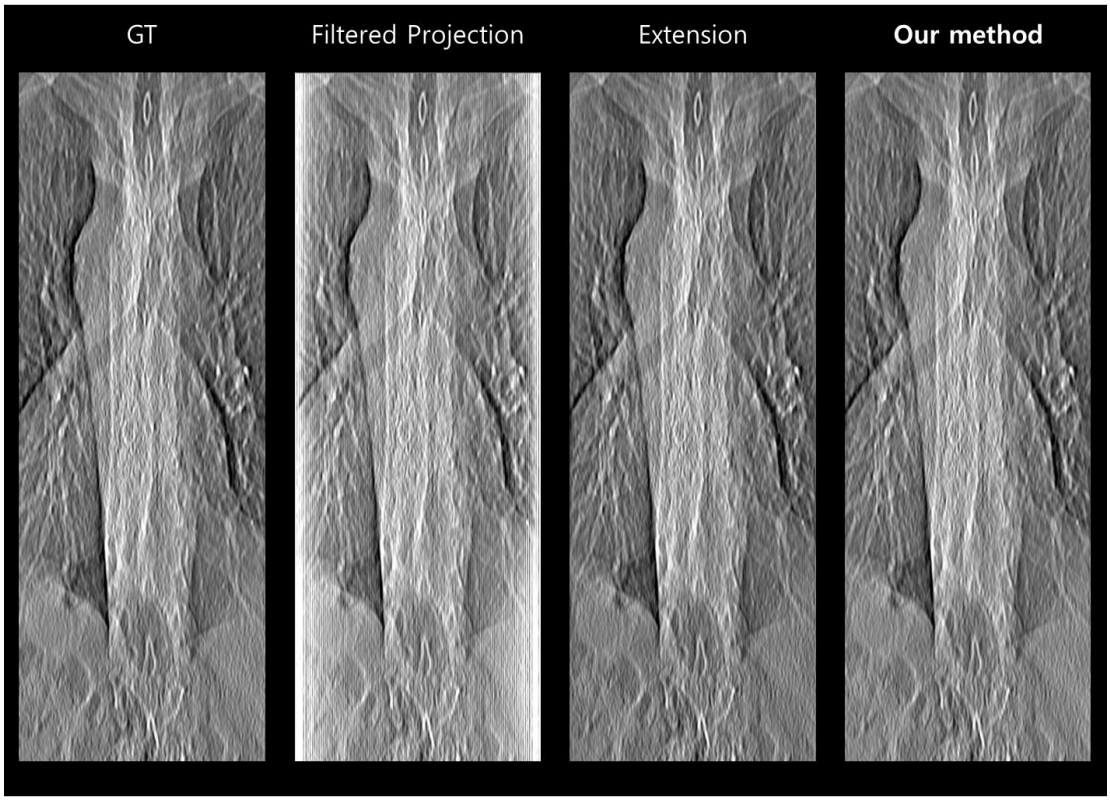
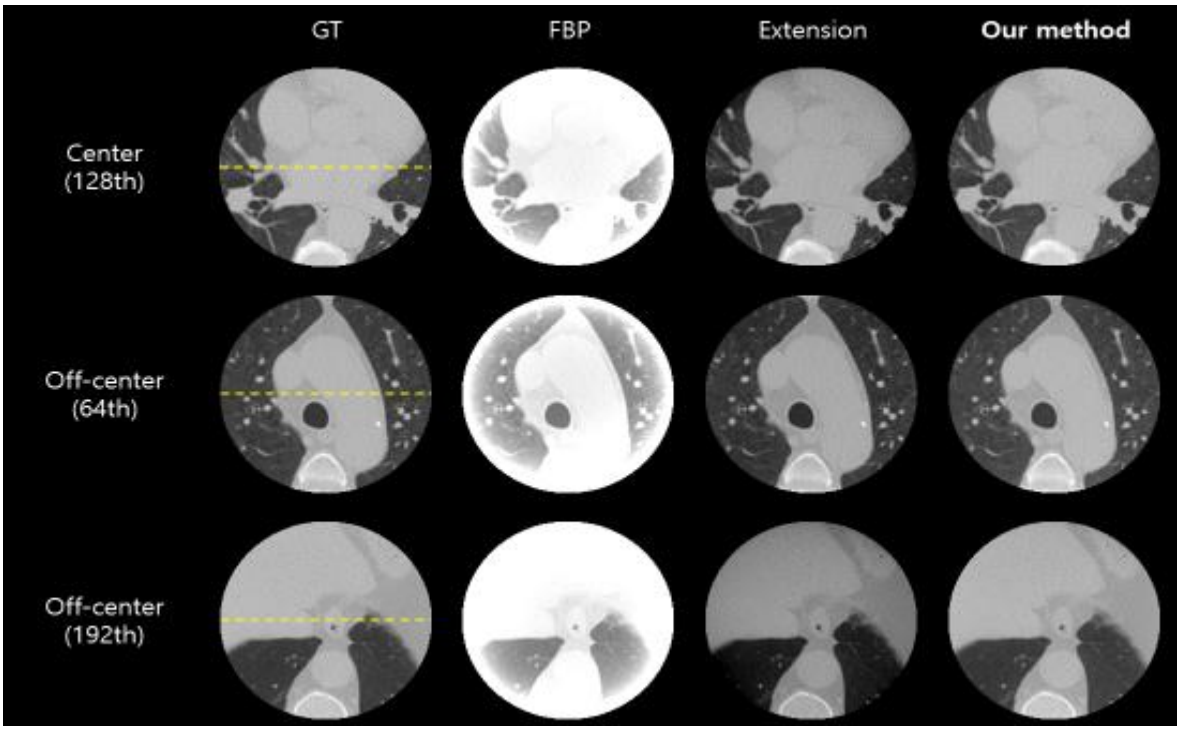


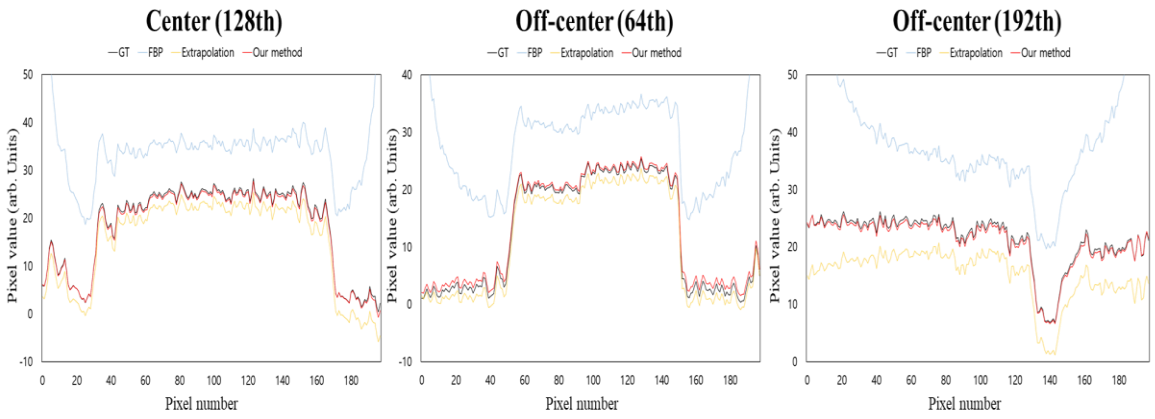
Figure 4. The figure shows images generated from GT, filtered projection, extension projection, and the proposed reconstruction method, respectively.

Also, Figure 5 represents the image reconstruction result of the proposed method of truncation artifact reduction. For comparison with this proposed method, GT, FBP method, and Extension method are shown together. As a result, the proposed ROI reconstruction method showed image quality very similar to that of GT. It was shown that not only the structure and shape but also the texture of the image was similarly corrected. In addition, it was shown that the truncation artifact was reduced to the same level not only in the central slice but also in the off-central slices.

The Root-Mean-Squared Error (RMSE) values for the test set of 14 patients were calculated 32.007, 3.544, and 0.927 in FBP, Extension, and the proposed ROI method, respectively. In addition, the Structural Similarity Index (SSIM) values were calculated 0.541, 0.709, and 0.979, respectively. Therefore, it can be seen that the proposed ROI reconstruction method excellently reduces artifacts and restores the image.



(a)



(b)

Figure 5. (a) Reconstructed images displayed: GT, FBP, Extension, and proposed method. (b) Comparison of mid-line profiles of the reconstructed images are shown by the GT (black), FBP (blue), Extension (orange), and our method (red). (WL: 14.5, WW: 45.5)

4. CONCLUSIONS

In this study, we developed a method to effectively reduce truncation artifacts in CBCT images. In particular, the proposed method devised a new method for reducing the artifacts in the filtering image domain different from the existing sinogram or reconstruction image domain[7,8]. The study was conducted using the chest, which is relatively difficult to extract features in deep learning, and it was confirmed that the performance was similar or higher than that of other papers.

As a result, the reconstructed ROI image quality and quantitative analysis results showed that the structure, shape, and image texture were similarly reconstructed compared to GT. We have achieved promising results and believe that the proposed ROI image reconstruction method can help reduce radiation dose to the patients while preserving image quality for applications such as image guided radiation therapy (IGRT), C-arm CT, etc. In the future, we plan to try hyper-parameter optimization and application to various parts of the human body.

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